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DRAWING API · BACKDROP

Use the Paint Editor to draw a brown dirt backdrop with 6 dark oval holes

📜 SCRIPT REFERENCE

Draw the Mole Field

STEPS TO FOLLOW

1. Open Scratch → delete cat sprite
2. Click backdrop thumbnail → "Paint" (pencil icon)
3. Fill canvas brown
4. Ellipse tool dark brown → draw 6 ovals in 3×2 grid
5. Click back-arrow to return

💡 TIP

Keep your hole ovals consistent in size — you'll position mole clones to match each hole centre in later steps.

🤔 DID YOU KNOW?

You just used a 2D drawing API — the same API that browsers use for HTML Canvas, and that Figma uses for its design canvas.

OBJECT TEMPLATES

Add a hedgehog or mole sprite, size it to 50, and hide it on flag click

SCRATCH BLOCKS — ON THE MOLE SPRITE

– MOLE: when 🏁 clicked →

```
hide
```

💡 TIP

If you can still see the mole, make sure the Code tab is on the MOLE sprite (blue outline in sprite panel).

🤔 DID YOU KNOW?

A hidden sprite with "when I start as a clone" scripts is exactly like a class — it defines behaviour without doing anything visible until instantiated.

RANDOM POSITIONING · CLONES

Make the mole teleport to a random hole position and create one clone

📜 SCRIPT REFERENCE

Pick a Random Hole

SCRATCH BLOCKS — ON THE MOLE SPRITE

– MOLE master: when 🏁 clicked →

```
hide
go to x: (pick random (-160) to (160)) y: (-60)
create clone of [myself]
```

💡 TIP

Adjust the x range to match your painted holes. If holes are at x=-120, 0, 120 use pick random (-120) to (120).

🤔 DID YOU KNOW?

"pick random" is a pseudo-random number generator — it produces numbers that feel random but are actually deterministic sequences computed from a seed.

TIMER · OBJECT LIFECYCLE

The mole clone appears, waits 1.5 seconds, hides, waits 0.5 seconds, then deletes itself

SCRATCH BLOCKS — ON THE MOLE SPRITE

— MOLE: when I start as a clone →

```
show
wait (1.5) seconds
hide
wait (0.5) seconds
delete this clone
```

💡 TIP

Try wait (0.8) for a faster pop-up or wait (2.5) for a slower, easier game. This timer controls difficulty!

🤔 DID YOU KNOW?

This show → wait → hide → delete pattern is the game-object lifecycle: create → active → deactivate → destroy. Used in every game engine for temporary objects.

Whack-a-Mole

INTERMEDIATE

MOUSE EVENTS · INPUT

Add a click handler on the Mole that plays a sound when whacked

SCRATCH BLOCKS — ON THE MOLE SPRITE

– MOLE: when this sprite clicked →

```
play sound [pop] until done
```

💡 TIP

Click the Sounds tab on the Mole sprite to browse and add sound effects. "Chomp", "Zap", or "Pop" all work well.

🤔 DID YOU KNOW?

"when this sprite clicked" is addEventListener("click") in JavaScript — the event pattern that powers every button, menu, and interactive element on the entire web.

VARIABLES · REWARD STATE

Create a Score variable and increment it on each successful click

SCRATCH BLOCKS — ON THE MOLE SPRITE

– MOLE: when this sprite clicked →

```
play sound [pop] until done
change [Score] by (1)
delete this clone
```

💡 TIP

Right-click the Score display and choose "Large readout" for a satisfying big counter.

🤔 DID YOU KNOW?

Every leaderboard, XP system, in-game currency, and achievement tracker is built on this same pattern: event detected → variable updated → display refreshed.

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PARALLEL EXECUTION · CONCURRENCY

Make the master mole spawn new clones continuously in a forever loop

SCRIPT REFERENCE

Moles Keep Coming

SCRATCH BLOCKS — ON THE MOLE SPRITE

– MOLE master: when 🏁 clicked →

```
hide
set [Score] to (0)
forever
  wait (1.2) seconds
  go to x: (pick random (-160) to (160)) y: (-60)
  create clone of [myself]
```

💡 TIP

Lower the wait time (0.8 secs) for more moles at once. Higher (2 secs) gives beginners more breathing room.

🤔 DID YOU KNOW?

Two "when 🏁 clicked" scripts running at the same time is concurrency — the same concept behind async/await in JavaScript and Python's threading module.

COUNTDOWN TIMER

Add a 30-second countdown Timer variable that ticks down once per second

SCRATCH BLOCKS — TIMER SCRIPT

when 🏁 clicked (timer script)

```
set [Timer] to (30)
repeat (30)
    wait (1) seconds
    change [Timer] by (-1)
```

💡 TIP

Increase to 45 or 60 seconds for an easier game. The "repeat" count always matches the start Timer value.

🤔 DID YOU KNOW?

This repeat-wait-change pattern is how every countdown works in code — from quiz timers to rocket launch sequences.

🏁

STATE MACHINES · PROGRAM TERMINATION

When the timer hits zero, show a message and stop the game

📜 SCRIPT REFERENCE

SCRATCH BLOCKS — AFTER THE REPEAT LOOP

(after the repeat loop)

```
say [Time's up! 🔨] for (2) secs
stop [all]
```

💡 TIP

Share your game on Scratch — click "Share" at the top. Friends can try to beat your high score!

🤔 DID YOU KNOW?

PLAYING → GAME_OVER is a state machine. Login flows, vending machines, traffic lights, and payment systems all work as state machines.